MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)
MG4DSCMAT200-MATRIX ALGEBRA AND NUMBER THEORY

Duration: 2 hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Write the 3×3 matrix whose ij^{th} entry is given by $x_{ij} = 2^{i+j}$. [U][1]
2. Is the product of two $n \times n$ diagonal matrices a diagonal matrix? Justify. [U][1]
3. Determine whether the rows of the matrix $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$ are linearly independent? [An][4]
4. Define an orthogonal matrix. [K][4]
5. State Euclid's Lemma. [K][5]
6. If the product of two positive integers a and b is 864 and $lcm(a, b)$ is 72, then find $gcd(a, b)$. [U][5]
7. Define pseudoprime. Give an example. [K][6]
8. Determine whether $20^{12} \equiv 1 (mod\, 13)$. [U][6]

Part B
Answer any five questions. Each question carries 6 marks

9. Prove that for every $m \times n$ matrix A, there exists a unique $m \times n$ matrix M such that $A + M = A$. [U][1]
10. Write the matrix representation of the following system of linear equations: [A][2]

$$2x + 3w - 2y + z - 5t = 9$$

$$-x + 2y + 7z - t = 8$$

$$4y + 8w - 2z = 7$$
$$2x + 2w - 4y + 3t = 6$$

11. Consider the matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 & 4 \\ 1 & \lambda & 1 & 1 & 4 \\ 1 & 1 & \lambda & 3-\lambda & 6 \\ 2 & 2 & 2 & \lambda & 6 \end{bmatrix}$. Show that if $\lambda \neq 1, 2$, then the row rank of A is 4.What is the row rank of A when $\lambda = 1$, and when $\lambda = 2$? [A][4]

12. Show that the system of equations

$$x + 2y + 3z + 3t = 3$$
$$x + 2y + 3t = 1$$
$$x + z + t = 3$$
$$x + y + z + 2t = 1$$

has no solution. [A][4]

13. Show that the expression $\frac{a(a^2+2)}{3}$ is an integer for all $a \geq 1$. [U][5]

14. Prove that for any integer a, one of the integers $a, a+2, a+4$ is divisible by 3. [U][5]

15. If p is a prime, then prove that $\phi(p^k) = p^k - p^{k-1}$, for any $k > 0$. [U][8]

16. Using Euler's theorem prove that for any integer a, $a^{37} \equiv a(mod\,1729)$. [U][8]

Part C
Answer any three questions. Each question carries 10 marks

17. (a) Reduce the matrix $\begin{bmatrix} 1 & 2 & 0 & 3 & 1 \\ 1 & 2 & 3 & 3 & 3 \\ 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 1 & 2 & 1 \end{bmatrix}$ to row-echelon form. [A][3]

(b) Reduce the matrix $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$ to Hermite form. [A][3]

18. Find the normal form of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & -2 \\ -1 & -1 & 1 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix}$. [A][4]

19. Define an invertible matrix. Determine whether the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 4 \end{bmatrix}$ is invertible? If so, find A^{-1}. [An][4]

20. (a) If p is a prime and $p|ab$, then show that $p|a$ or $p|b$. [U][5]
(b) Prove that the number $\sqrt{2}$ is irrational. [U][5]

21. (a) Show that $41|2^{20}-1$. [U][5]
(b) Find the remainder obtained upon dividing the sum $1!+2!+3!+\cdots+100!$ by 12. [U][5]

22. If p is a prime, prove that $(p-1)! \equiv -1(mod\, p)$. [U][7]